



Lecture 1: Cyber Security: Introduction

CYS401: Fundamental of Cyber Security

Outline

- ❑ What is security?
- ❑ Critical characteristics of information,
- ❑ Components of an Information System,
- ❑ The system development life cycle,
- ❑ The security system development life cycle.

ICE-BREAKING

About you.

Simple quiz on cybersecurity

<https://www.pewresearch.org/internet/quiz/cybersecurity-knowledge/>



Introduction

Cyber security is:

- the protection of internet-connected systems, including hardware, software and data, from cyberattacks.

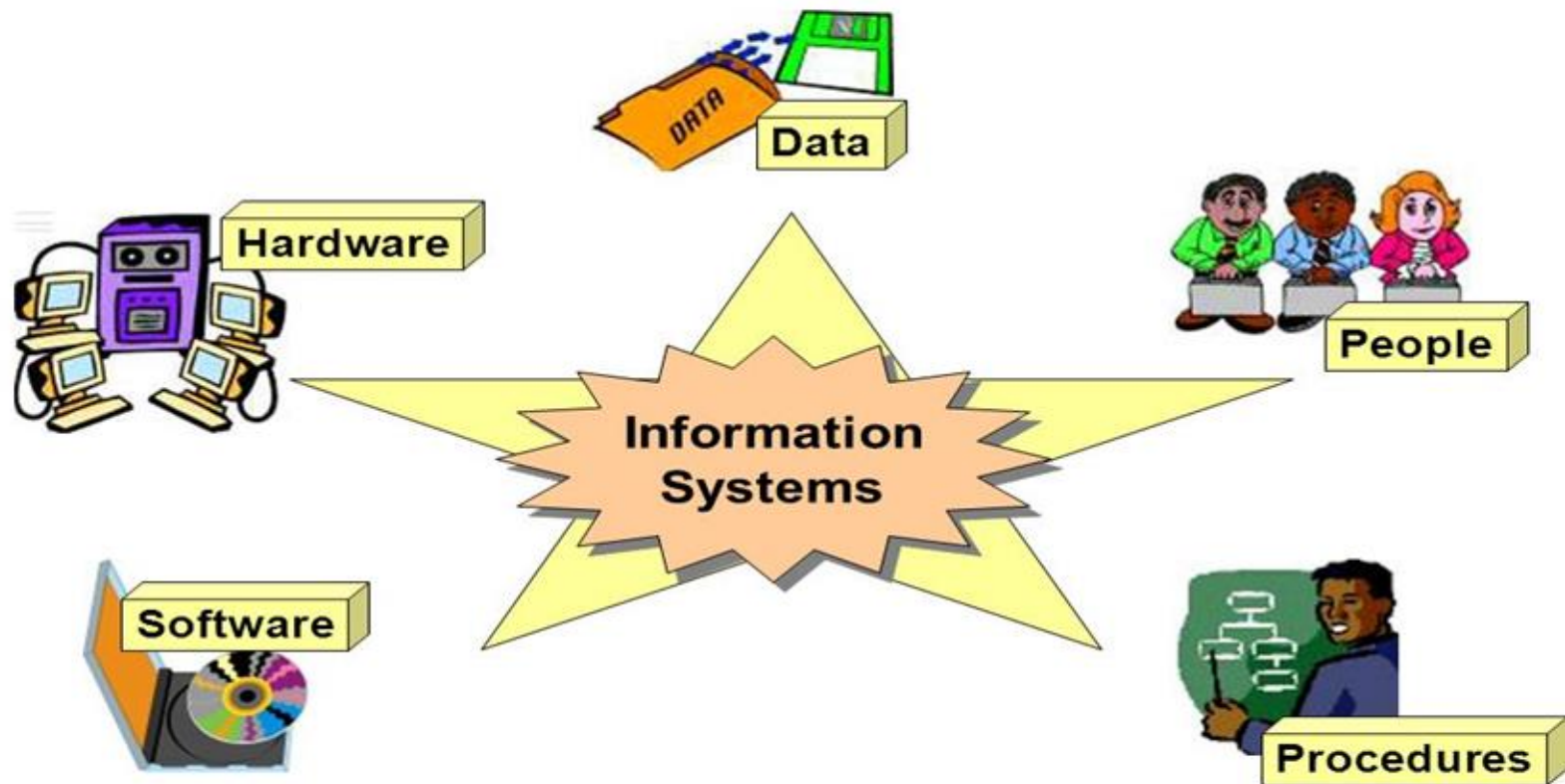


GETTING STARTED!



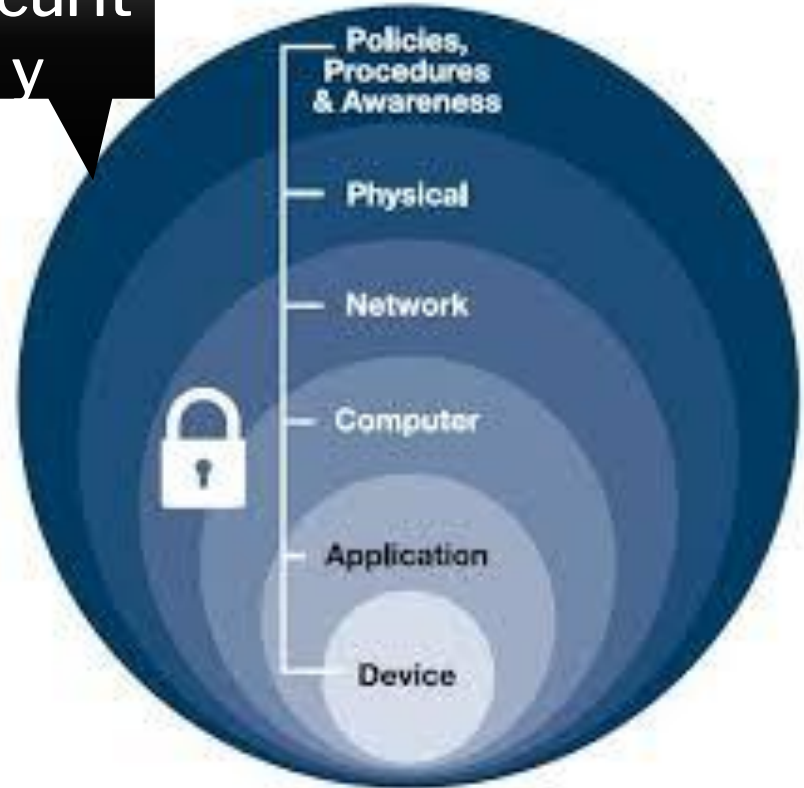
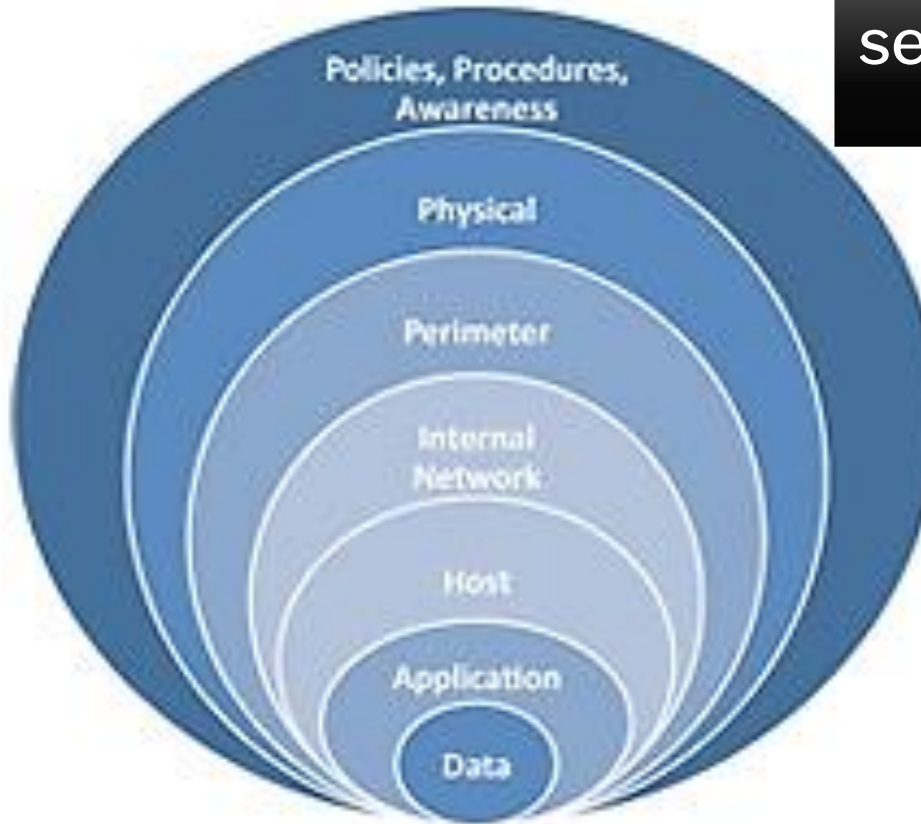
Introduction

5 Components of an IS



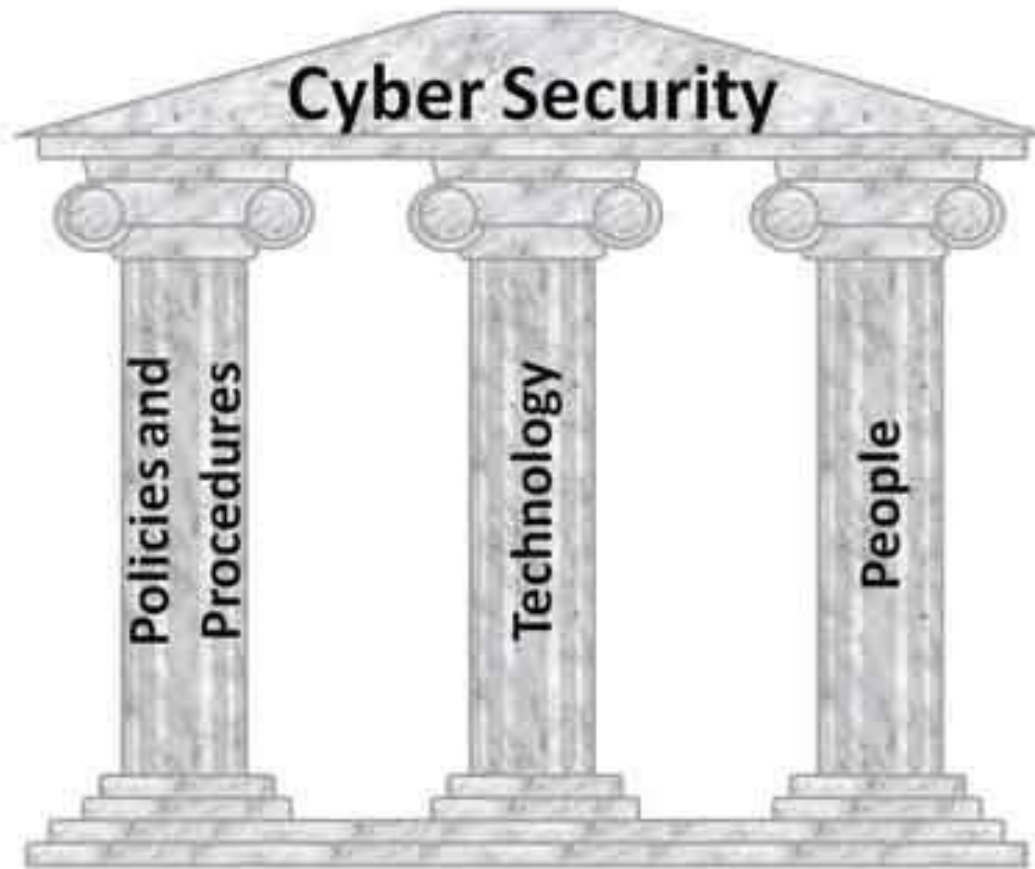
Introduction

Cyber
security



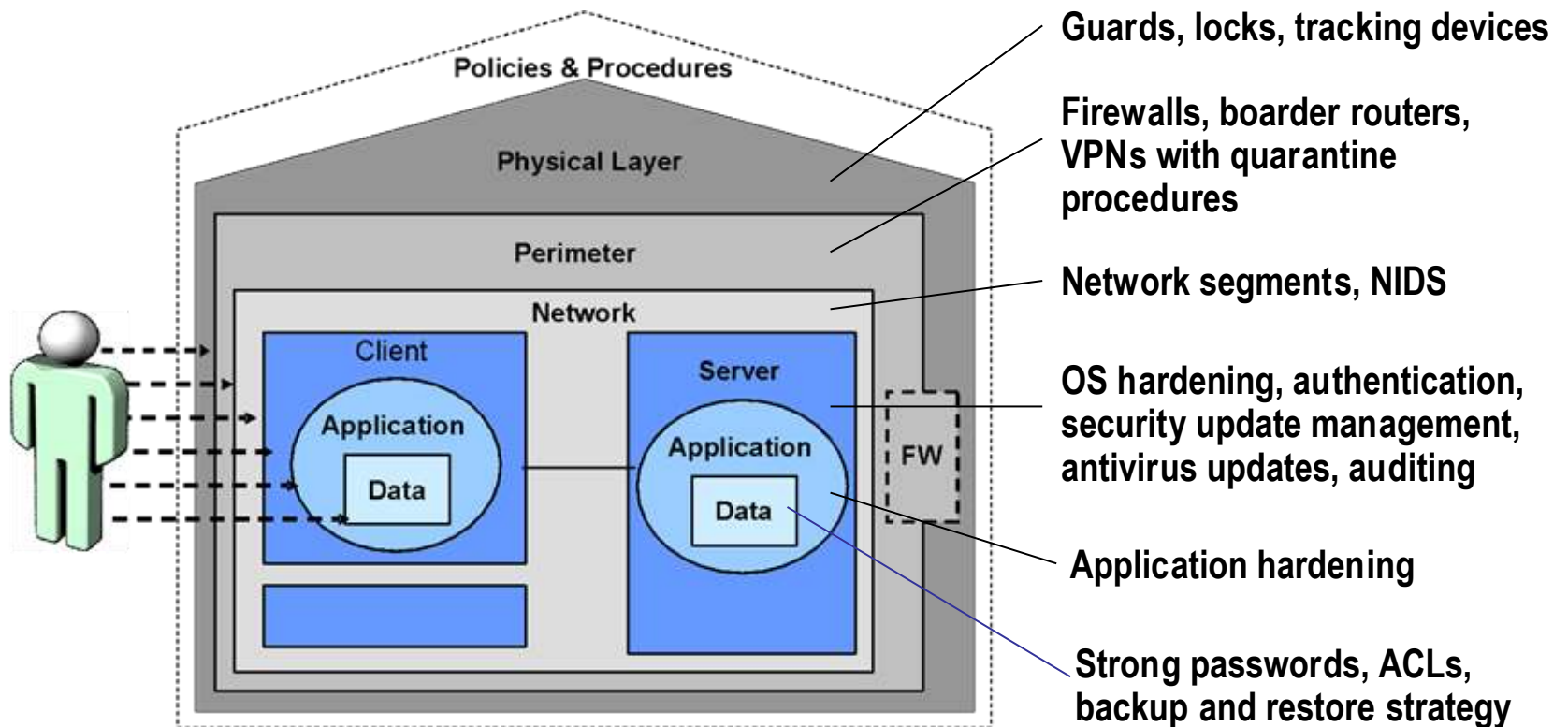
Introduction

- Cybersecurity pillars:



Introduction

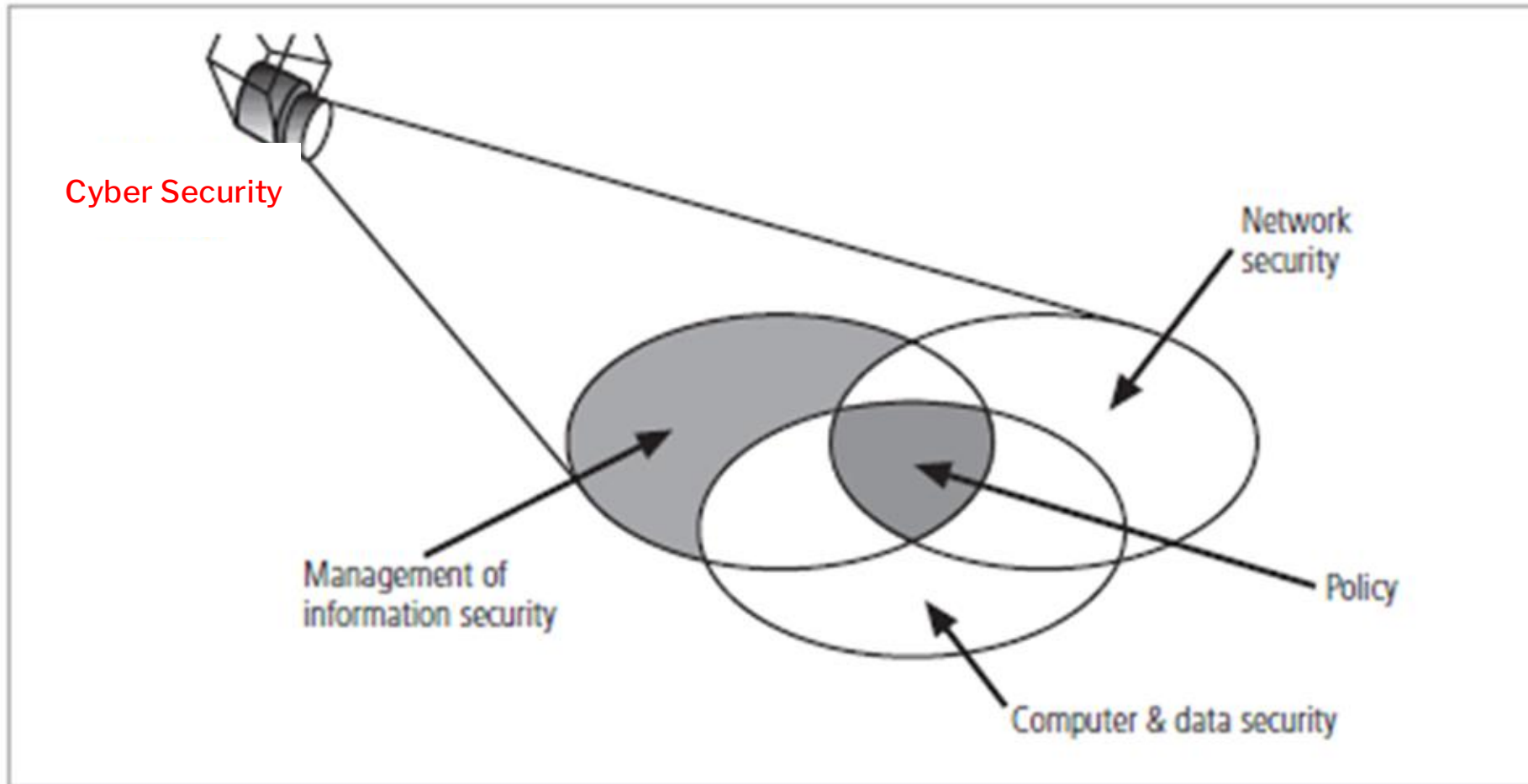
Cyber security aims to build a **defense-in-depth approach**



Introduction

- ❑ A successful organization should have a multi-layer of security in place.
 - ❑ **Physical security**
 - ❑ Secure **physical items, objects** – from unauthorized access and misuse
 - ❑ **Personnel security**
 - ❑ Protect the **individual(s)** who are authorize to access the organization and its operations
 - ❑ **Operations security**
 - ❑ Protect the details of a particular **operations / activities**
 - ❑ **Communications security**
 - ❑ Protect **communications media / technology**
 - ❑ **Network security**
 - ❑ Protect **network components / connection**
 - ❑ **Data security**
 - ❑ Protects **Confidentiality / Integrity and Availability of information** assets – storage / processing or even during transmission

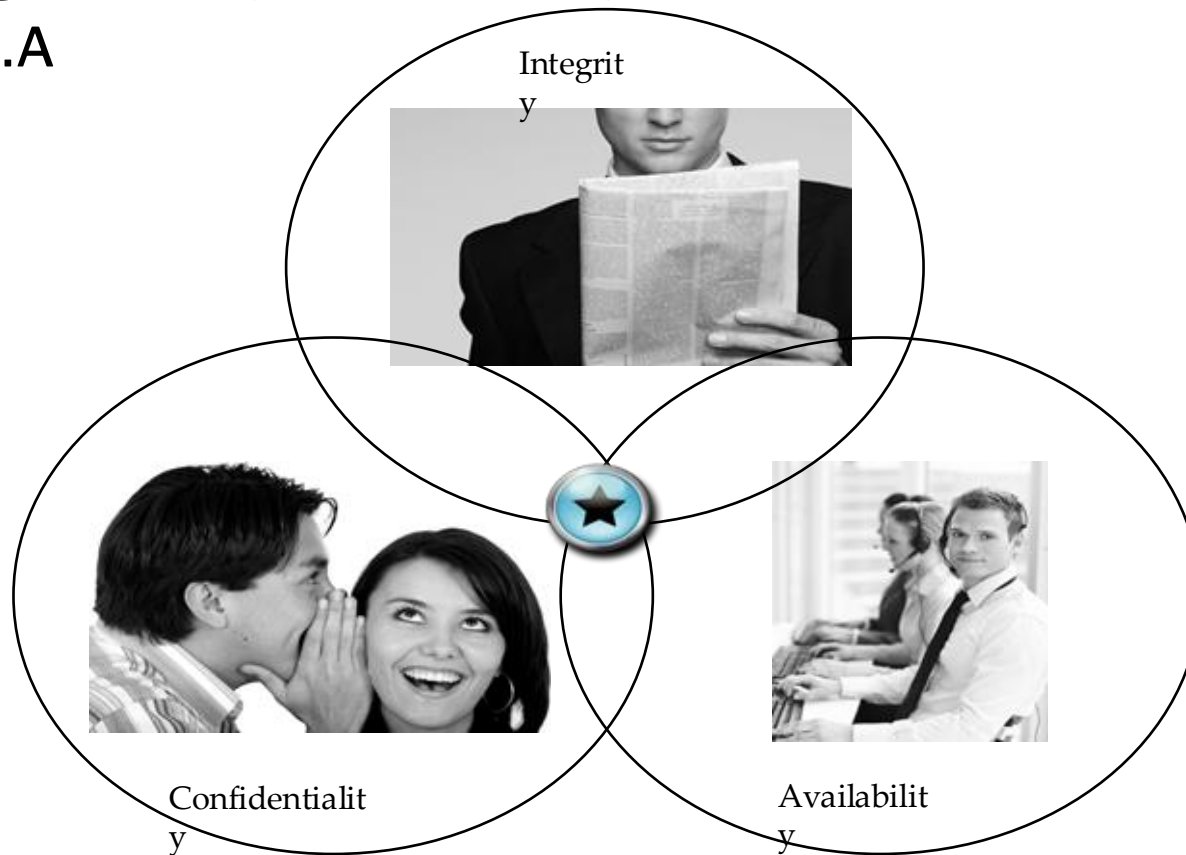
Introduction



Critical characteristics of information

❑ The goal of CyberSecurity?

❑ C.I.A



Critical characteristics of information

- ❑ C.I.A are the critical characteristics of Information, but there are other characteristics should be considered.
 - ❑ Confidentiality
 - ❑ Integrity
 - ❑ Availability
 - ❑ Accuracy
 - ❑ Authenticity
 - ❑ Utility
 - ❑ Possession
- ❑ The value of information comes from the characteristics it possesses:

Critical characteristics of information

- ❑ **Confidentiality** – The quality or state of preventing disclosure or exposure to unauthorized individuals or systems.
 - ❑ Credit Cards, PII, Health records.
- ❑ **Integrity** – The quality or state of being accurate, complete, and authorised.
 - ❑ The integrity of information is threatened when the information is exposed to corruption, damage, destruction, or other disruption of its authentic state.
- ❑ **Availability** – Enables authorized users who need to access information to do so without interference or obstruction.
 - ❑ The information is said to be available to an authorized user when and where needed and in the correct format.

Critical characteristics of information

- ❑ **Accuracy** – Free from mistakes or errors and having the value that the end user expects.
 - ❑ If information contains a value different from the user's expectations due to intentional or unintentional modification of the content, it is no longer accurate.
- ❑ **Authenticity** – The quality or state of being genuine or original, rather than a reproduction or fabrication.
 - ❑ Information is authentic when the information is the same as it was originally created, placed, stored, or transferred.
- ❑ **Utility** – The quality or state of having value for some purpose or end.
- ❑ Information has value when it serves a particular purpose. This means that if information is available, but not in a format meaningful to the end user, it is not useful.
- ❑ **Possession** – The quality or state of having ownership or control of some object or item.
- ❑ Information is said to be in possession if one obtains it, independent of format or other characteristic.

Critical characteristics of information

- ❑ While a breach of confidentiality always results in a breach of ownership, a breach of ownership does not always result in a breach of confidentiality.
- ❑ Is it correct, **how and why?**

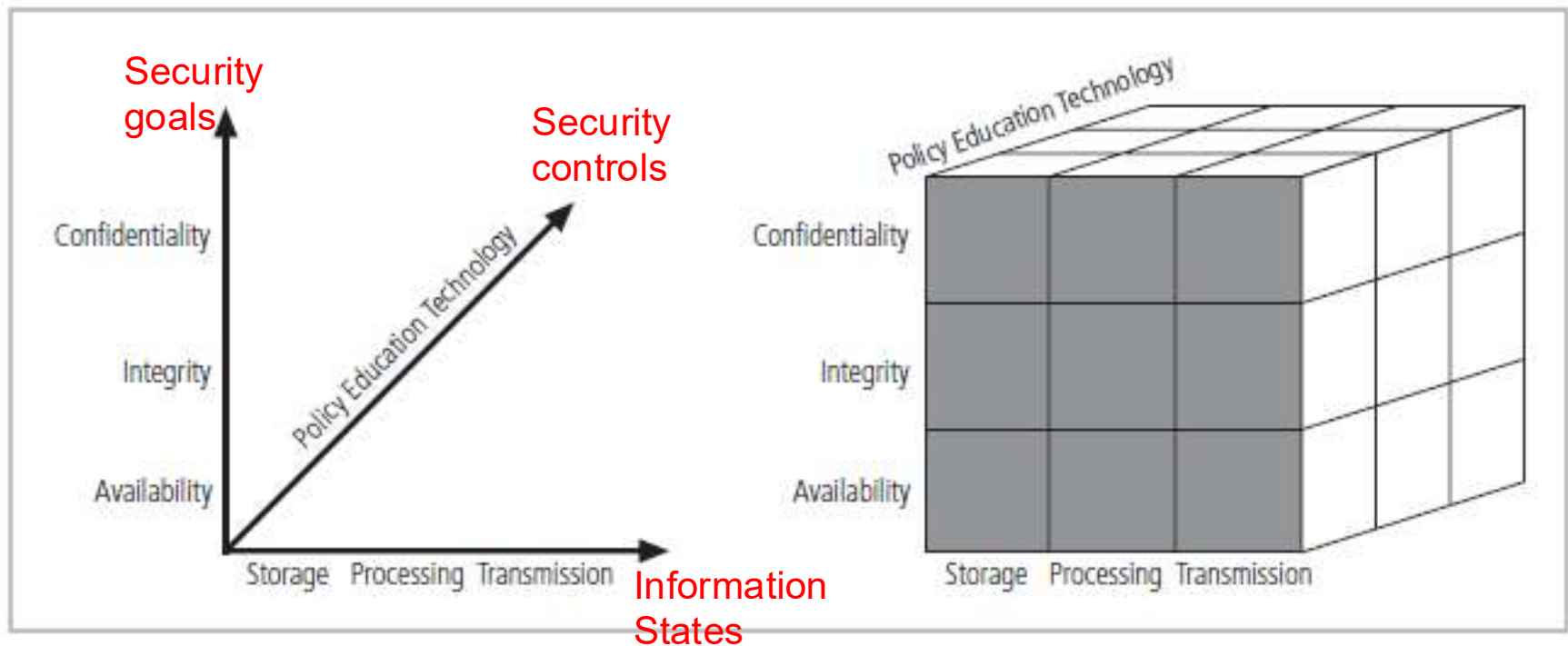


Critical characteristics of information

- In Cybersecurity, there are THREE dimensions
 - Security Goals: C.I.A.
 - Security measures/controls: policy, education/training, Technology.
 - Information States: Storage, Transmission, Processing
- McCumber Cube: graphical explanation to show the interconnections among the different Information security factors.

Critical characteristics of information

❑ McCumber Cube



Critical characteristics of information

- ❑ McCumber Cube: Dimensions and attributes
 - ❑ **Desired goals:** C.I.A
 - ❑ **Information states**
 - ❑ **Storage:** Data at rest (DAR) in an information system, such as that stored in memory or on a magnetic tape or disk.
 - ❑ **Transmission:** transferring data between information systems - also known as data in transit (DIT).
 - ❑ **Processing:** performing operations on data in order to achieve a desired objective.
 - ❑ **Security controls:**
 - ❑ **Policy and practices:** administrative controls (plans and guidance)
 - ❑ **Education:** ensuring that the users of information systems are aware of their roles and responsibilities.
 - ❑ **Technology:** software and hardware-based solutions designed to protect information systems.

Components of an Information System

- ❑ There are five components work together to achieve the ultimate goal of any information system.
 - ❑ Software
 - ❑ Hardware
 - ❑ Data
 - ❑ People
 - ❑ Procedures
- ❑ Agreeable to say that all components have their strengths and weaknesses – but when in used all together it performs a good secure IS

Components of an Information System

☐ Software

- ☐ Applications / operating systems / other utilities
- ☐ Most **difficult to secure**
- ☐ **Bugs / errors** are just too much to handle which leads to having a insecure information
- ☐ SW is the most important component in IS – but **information security is often to not being considered at the first round of implementation** – leads to having versions after versions of app.
- ☐ By this – it is an easy target to attack the IS

Components of an Information System

☐ Hardware

- ☐ Physical technology that houses and executes the SW/ stores and transmit the data and provides interfaces to run the SW
- ☐ Securing physical assets from harm / theft
 - ☐ Traditional and still mainly used – locks and keys
 - ☐ Biometrics access controls

☐ Data

- ☐ Most valuable asset possessed by an organization and often the main target of *intentional attacks*

Components of an Information System

☐ People

- ☐ User of the IS
- ☐ Often threat or being threat for the data
- ☐ Often being **the weakest component** in an IS environment
- ☐ Policy, agreement, education and training plays important rules

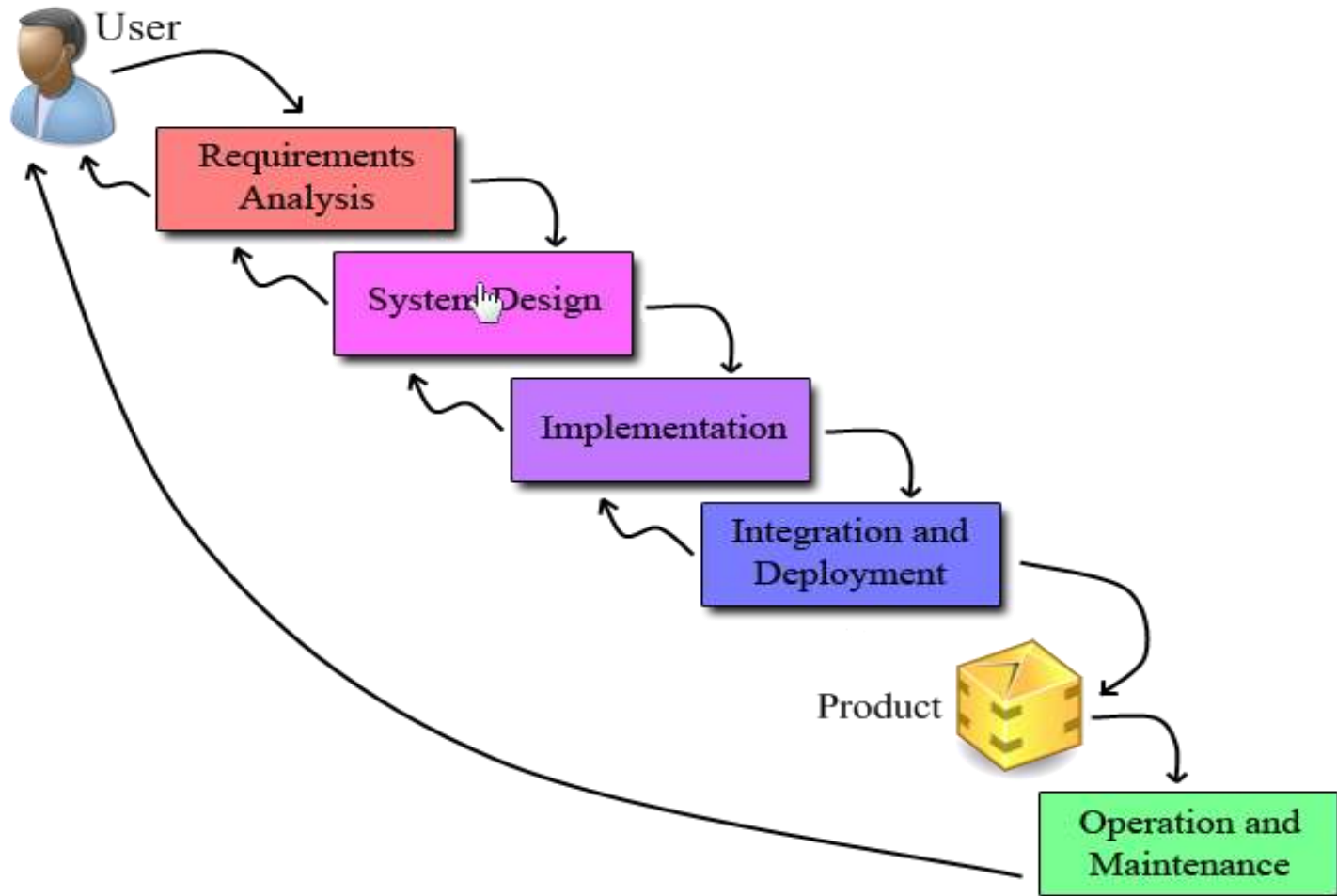
☐ Procedures

- ☐ Written instructions for accomplishing a specific task
- ☐ If a unauthorized user obtains an organization's procedure – this poses a threat to the integrity of the information
- ☐ Most organization distribute procedures to their trustful employees so they can access the IS

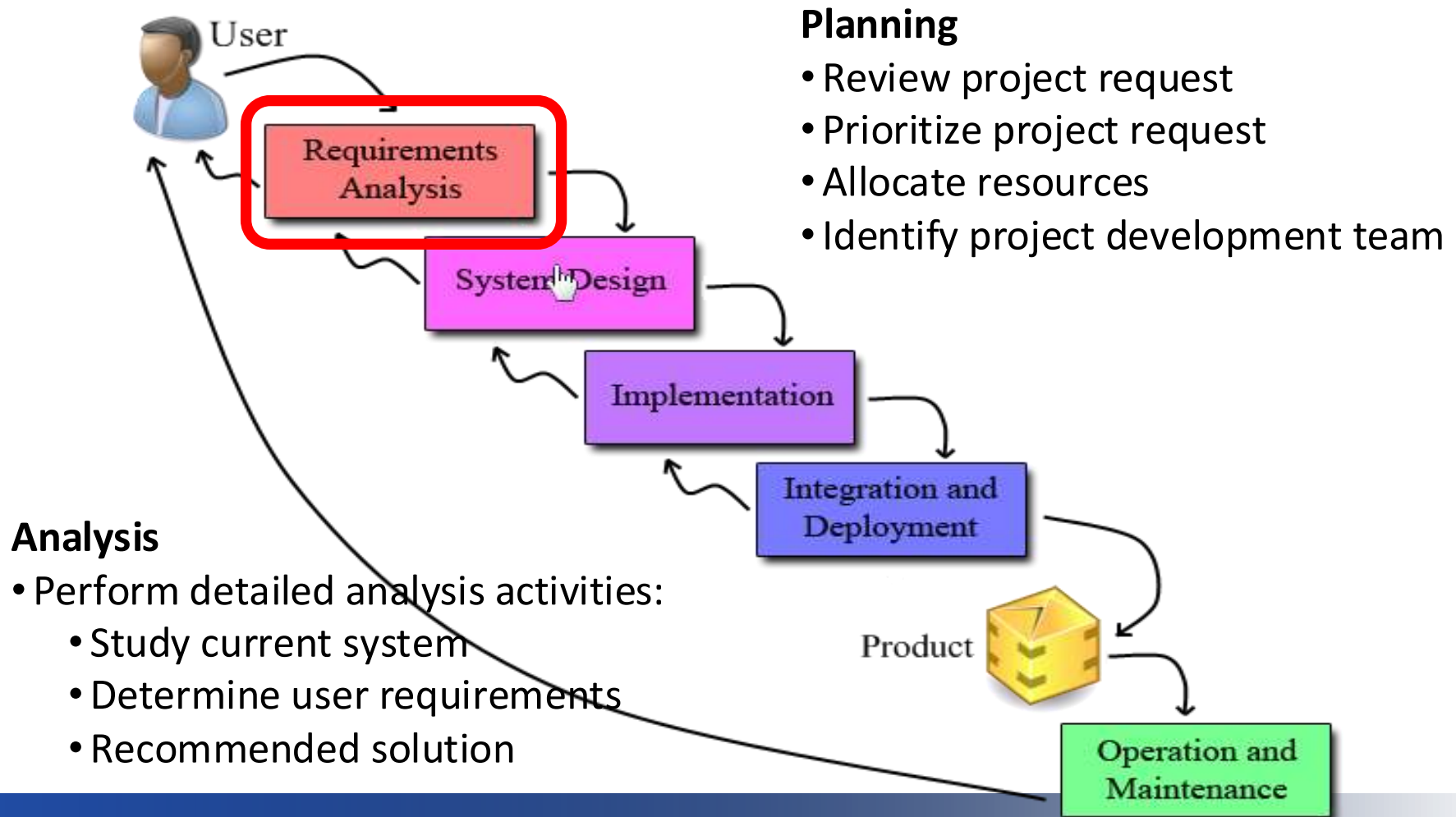
System Development Life Cycle (SDLC)

- Is a methodology / approach for designing and implementing an IS
- Sequence of procedure to solve a problem
- Most common approach – waterfall model

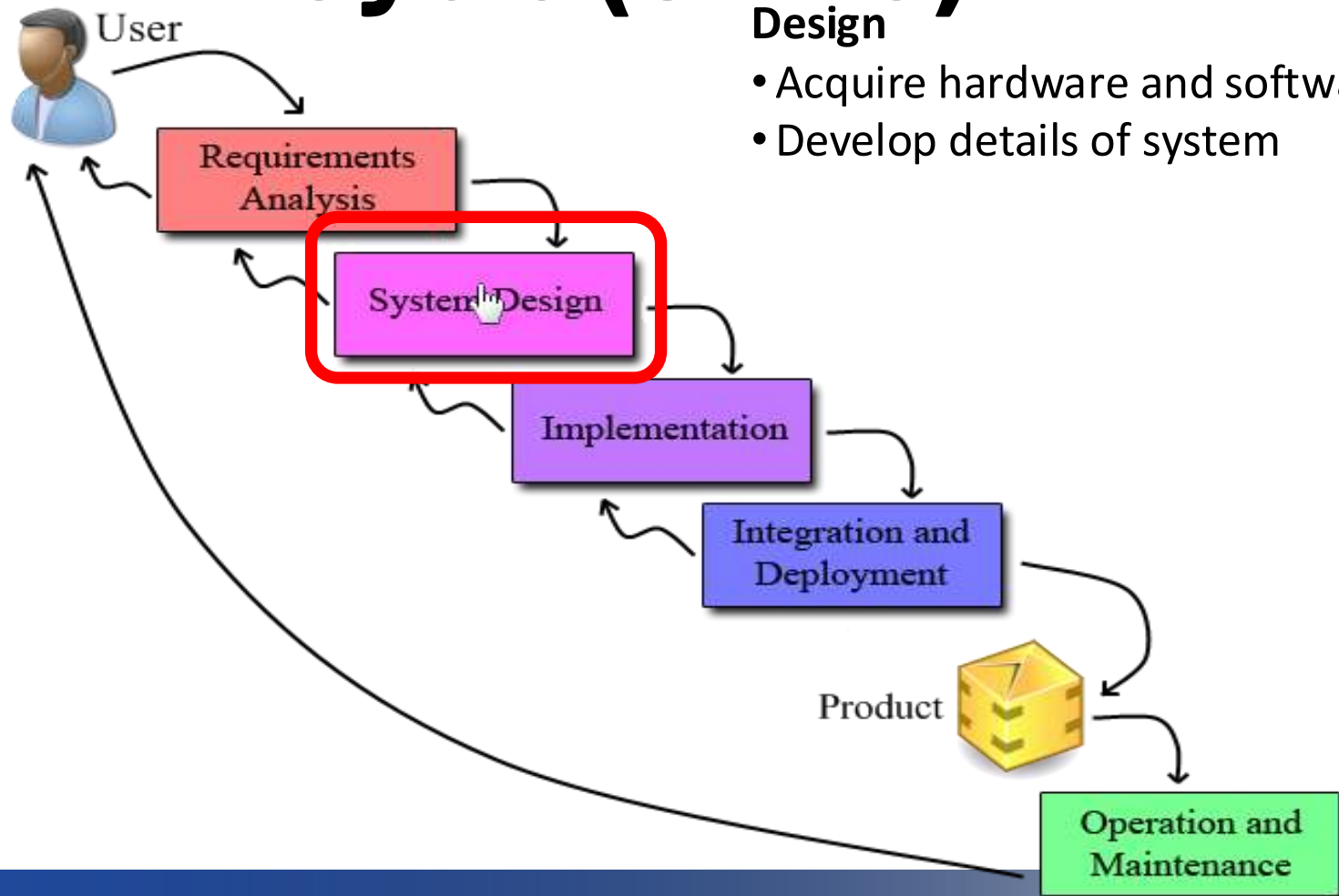
System Development Life Cycle (SDLC)



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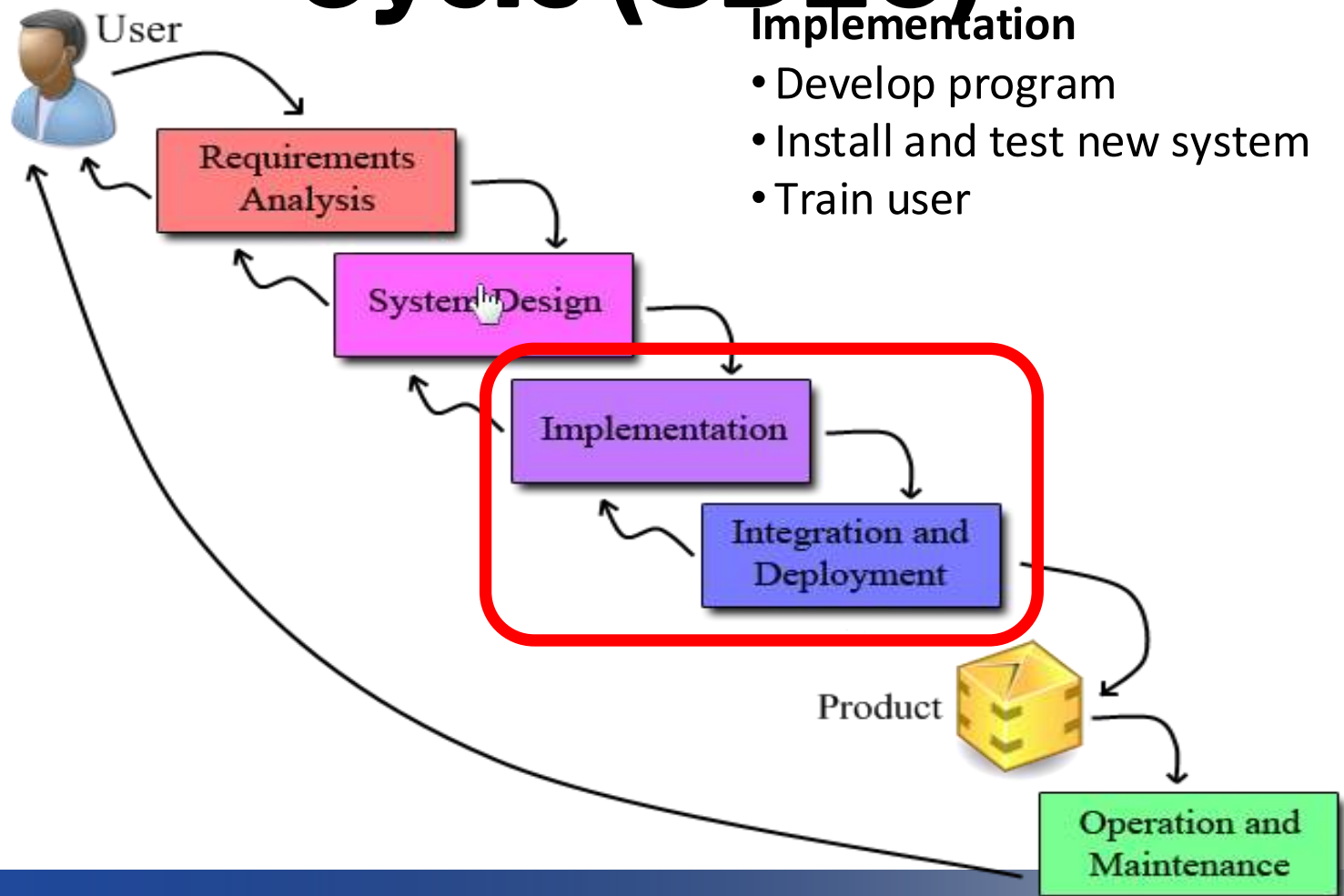
System Development Life Cycle (SDLC)



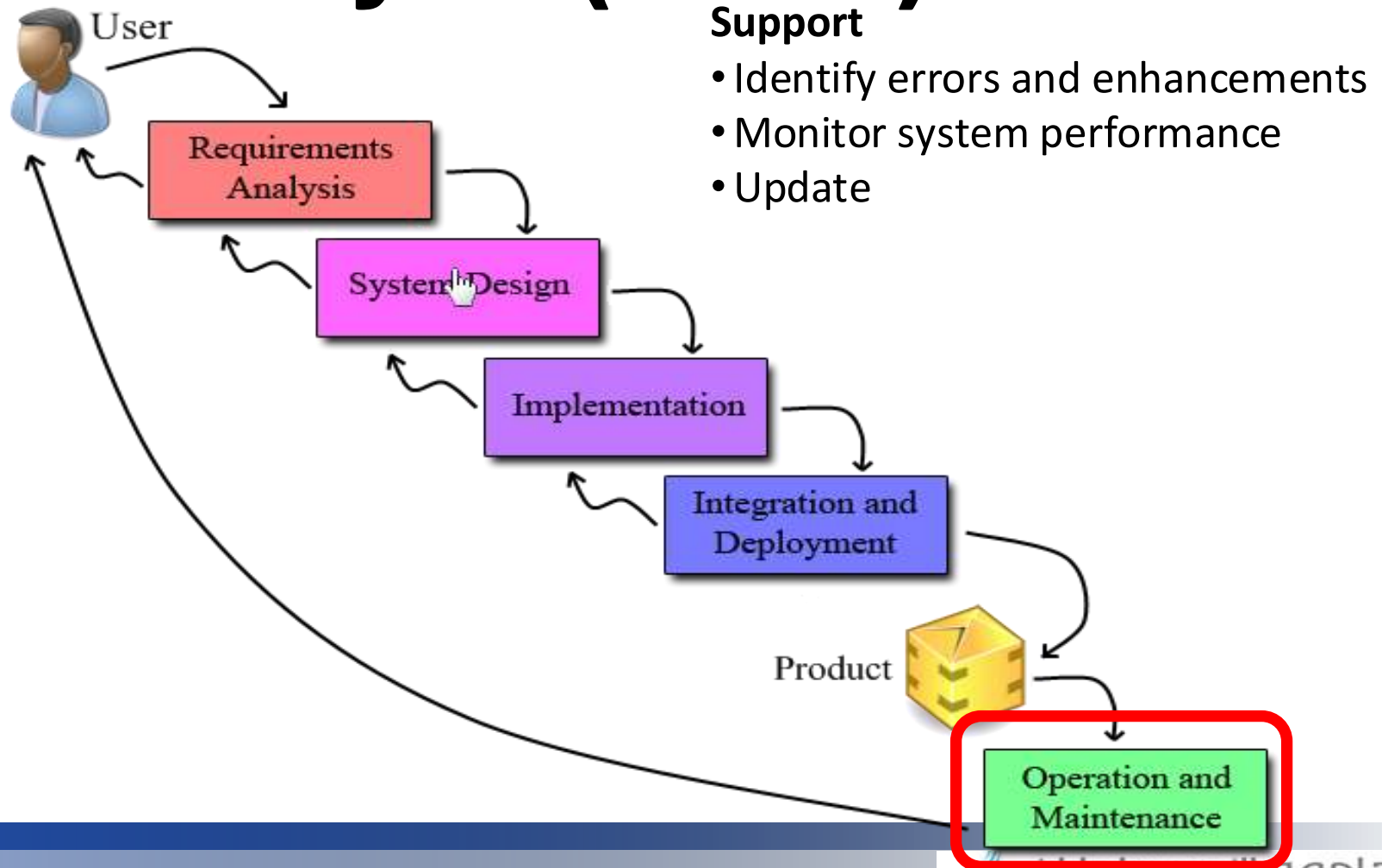
Design

- Acquire hardware and software
- Develop details of system

System Development Life Cycle (SDLC)



System Development Life Cycle (SDLC)



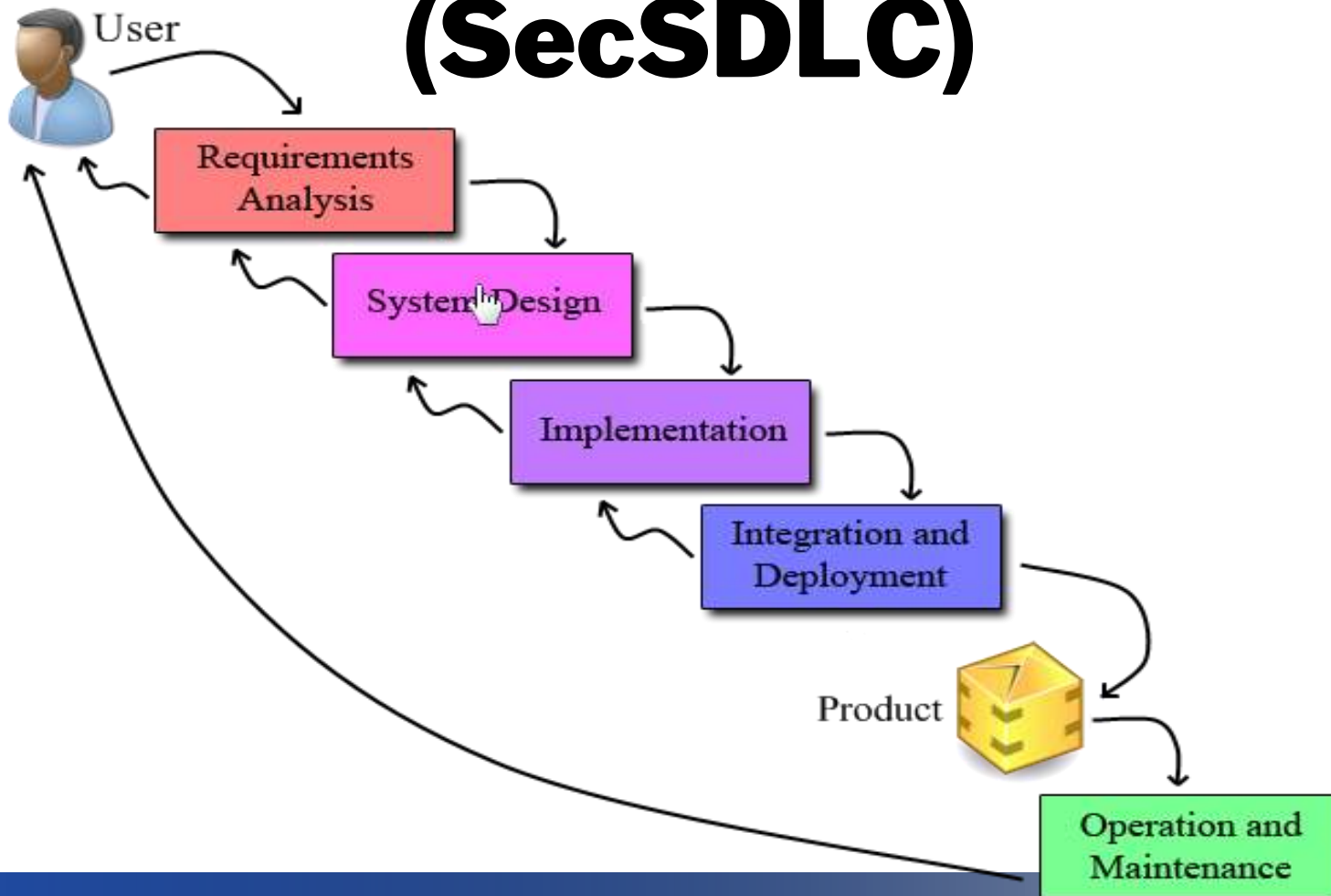
Security System Development Life Cycle (SecSDLC)

- Cybersecurity must be managed in a manner similar to any other system implementation
- One approach for implementing an cybersecurity system **with security in place** is to use a variation of the SDLC phases, named **Security SDLC** (SecSDLC)

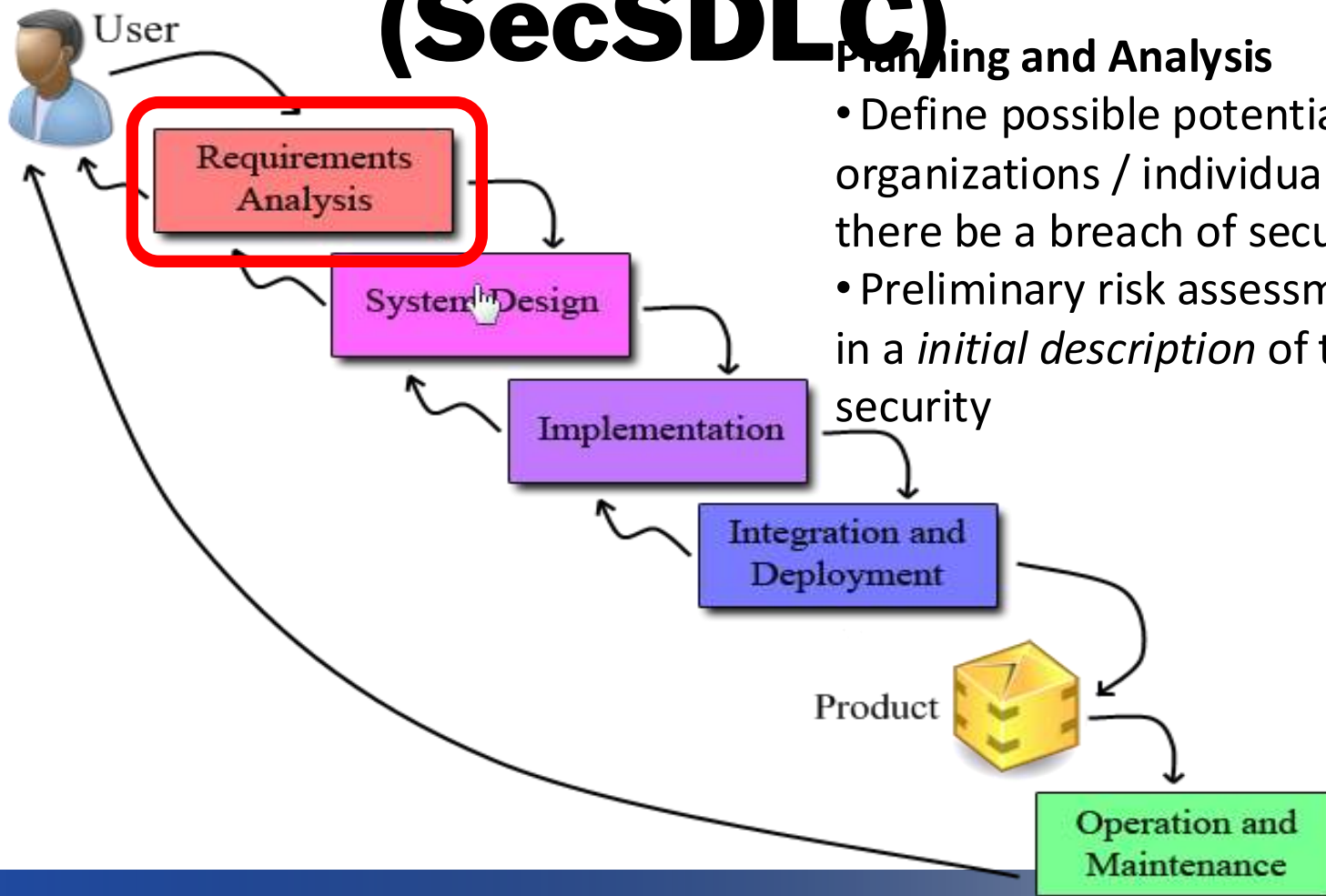
Security System Development Life Cycle (SecSDLC)

- Each phases in SecSDLC should consider the security of the system being assembled as well as the information it uses
- Each implementation done is secure and does not harm the C.I. A of the organization's information assets

Security System Development Life Cycle (SecSDLC)



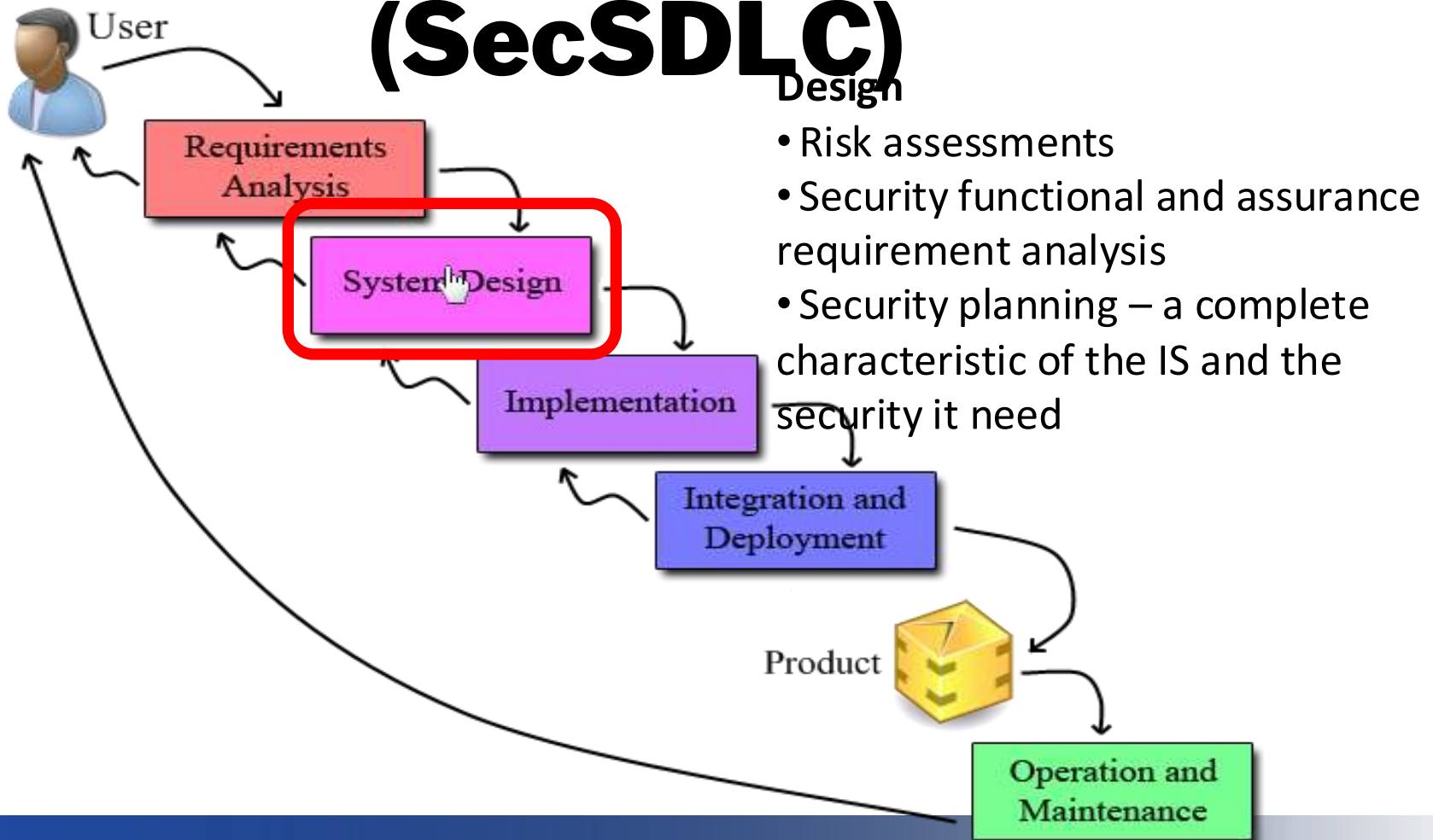
Security System Development Life Cycle (SecSDLC)



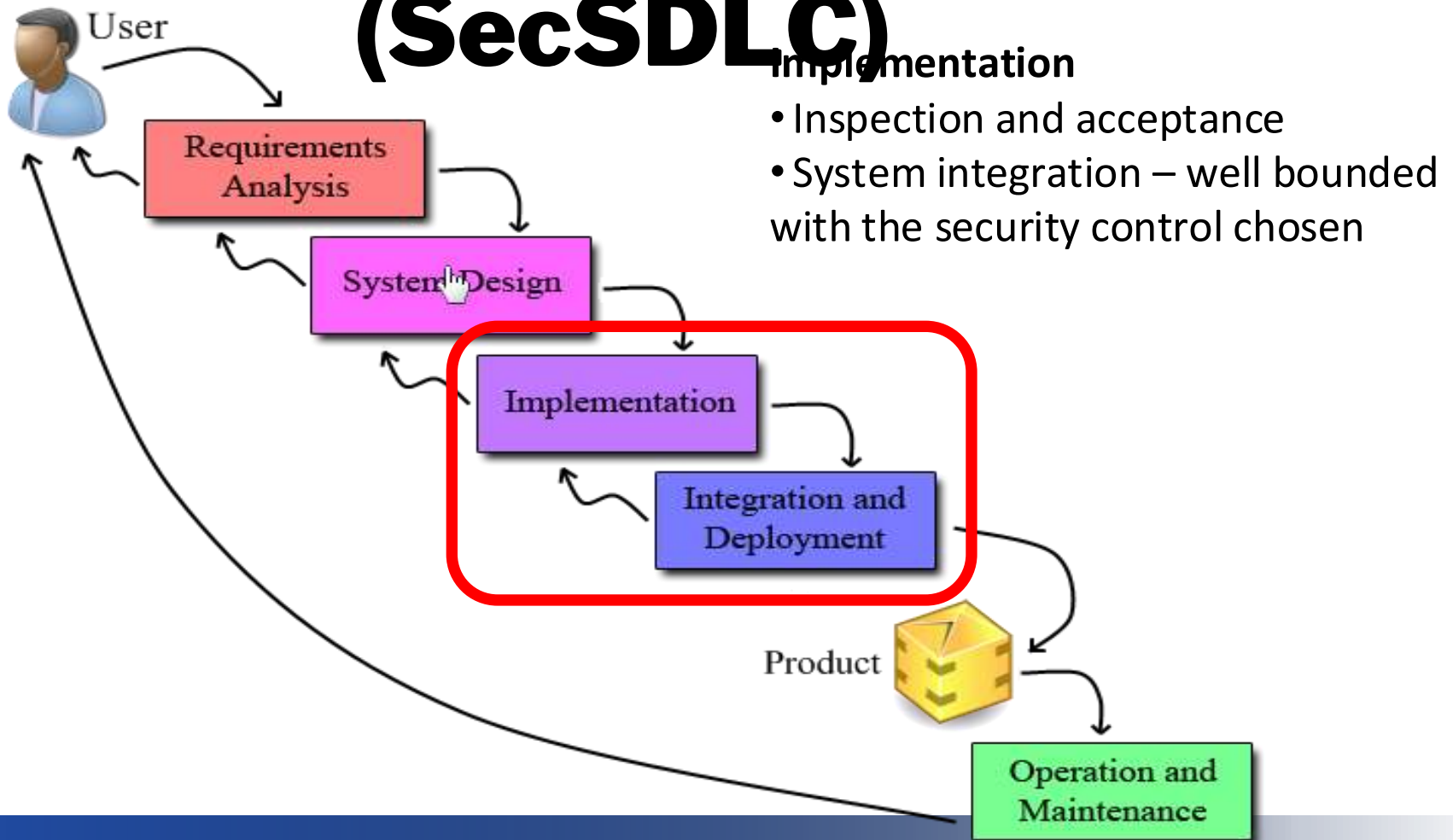
Planning and Analysis

- Define possible potential impact on organizations / individuals should there be a breach of security
- Preliminary risk assessment – result in a *initial description* of the basic security

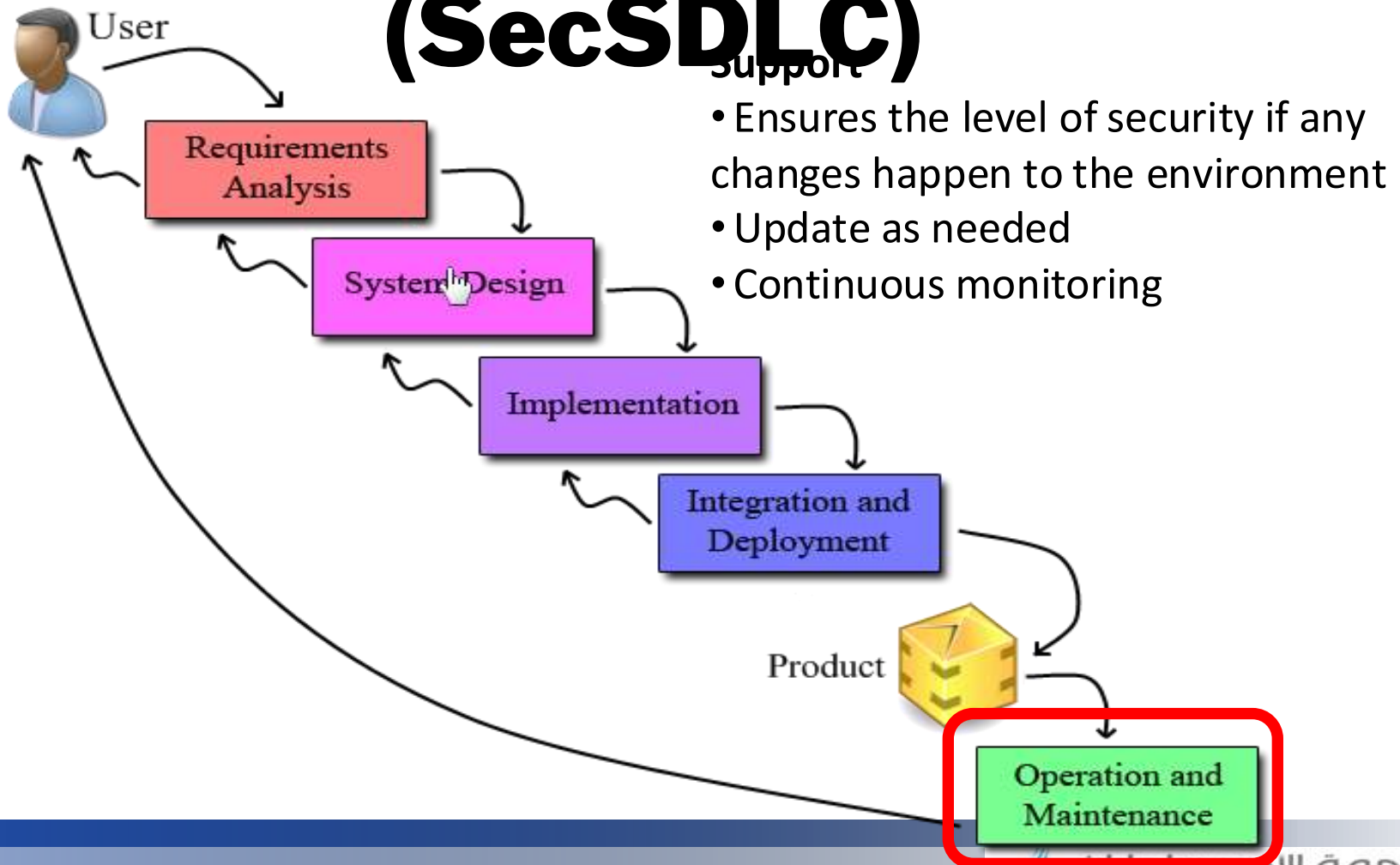
Security System Development Life Cycle (SecSDLC)



Security System Development Life Cycle (SecSDLC)



Security System Development Life Cycle (SecSDLC)



SDLC vs. SecSDLC

- ❑ SecSDLC is basically following the exact phases as what the formal SDLC is doing
- ❑ In addition, **SecSDLC identifies specific threats that possibly could happen to the IS and creates specific controls to counter those threats**
- ❑ A well-observe and planning IS could be demonstrated if all possible threats been taken care off

